

REPORT

1. Artificial Intelligence (AI) :is a branch of computer science that focuses on building machines or software systems that can perform tasks that normally require human intelligence.

In simple terms:

AI means making computers think and act like humans in certain situations.

These tasks include:

- **Learning** from data (like studying past examples)
- **Reasoning** (solving problems logically)
- **Understanding language** (like chatbots)
- **Recognizing images and speech**
- **Making decisions** (like recommendation systems in YouTube or Netflix)

Example:

- When Google Maps suggests the fastest route → AI is working.

1. Artificial Intelligence (AI)

Definition:

AI is the **broad field** of making machines behave like humans (thinking, reasoning, decision-making).

Examples:

- Chess-playing programs
- Google Maps route planning

2. Machine Learning (ML)

Definition:

ML is a **subset of AI** where machines learn patterns from data without being explicitly programmed.

Examples:

- Spam email detection
- Netflix recommendations
- **3. Generative AI (GenAI)**
- **Definition:**
Generative AI is a **subset of ML (especially deep learning)** that can **create new content** like text, images, audio, or code.
- **4. Agentic AI**
- **Definition:**
Agentic AI is an advanced type of AI that can **take actions autonomously to achieve a goal**, not just generate output.

Examples:

- AI that books tickets for you automatically
- AI coding agents that build full projects

3.1. BERT (Bidirectional Encoder Representations from Transformers)

BERT

What it is:

BERT is a **natural language processing (NLP) model** developed by Google that understands text by reading it in **both directions (left and right)**.

Key features:

- Understands context of words in a sentence
- Pre-trained on large text datasets
- Fine-tuned for tasks like classification

Use cases:

- Sentiment analysis (positive/negative reviews)
- Question answering systems
- Text classification (spam detection)

2. LLaMA (Large Language Model Meta AI)

LLaMA

What it is:

LLaMA is a **large language model (LLM)** developed by Meta that is designed to generate and understand human-like text.

Key features:

- Efficient and smaller compared to many LLMs
- Strong performance in reasoning and writing tasks
- Available in multiple versions (LLaMA 1, 2, 3)

Use cases:

- Chatbots and AI assistants
 - Code generation
 - Content writing and summarization
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3. Stable Diffusion

Stable Diffusion

What it is:

Stable Diffusion is a **generative AI model** that creates images from text descriptions.

Key features:

- Uses diffusion process to generate high-quality images
- Open-source and widely used in creative AI tools
- Runs on consumer GPUs

Use cases:

- AI art generation
- Advertising design
- Game and animation concept art

1. Traditional Development

Definition:

Traditional development is where software is built using **manual coding**, based on fixed requirements written by humans.

Key characteristics:

- Developers write **all code manually**
- Follows strict requirements and design documents
- Logic is explicitly programmed (if-else, loops, algorithms)
- No learning from data
- Changes require manual updates in code

Example:

- Building a calculator app using Java by writing every function yourself
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2. AI-Assisted Development

Definition:

AI-assisted development uses **AI tools and models to help developers write, debug, and optimize code faster and smarter.**

Key characteristics:

- AI helps generate or suggest code (e.g., GitHub Copilot, ChatGPT)
- Can auto-complete functions and detect bugs
- Uses data and context to improve productivity
- Supports natural language → code conversion
- Faster development cycle

Example:

- Writing: “create a login page in React” → AI generates code automatically

AI Hallucinations

- **Definition:**
AI hallucination is a situation where an AI system (like a chatbot or generative model) **produces incorrect, misleading, or completely made-up information**, even though it sounds confident and realistic.
- **Example of AI Hallucination:**
- **Question:**
- “Who won the 2023 Cricket World Cup?”
- **AI Hallucinated Answer:**
- “India won the 2023 Cricket World Cup by defeating Australia in the final.”
- **Reality:**
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- Australia won the 2023 Cricket World Cup.

